Mathematics – I (M-I)
GTU # 3110014

Dedicated Faculty, Committed Education
Darshan
Institute of Engineering & Technology

Unit – 5
Multiple integral

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Outline

- D.I. over general region – PC
- D.I. by Change of Order of integration
- D.I. by Change of Variable of integration – CC & PC
- Area by D.I. – CC & PC
- Volume by D.I.
- T.I. over General Region

METHOD – 6 : D.I. OVER GENERAL REGION IN POLAR COORDINATES(1)

METHOD – 5 : D.I. BY CHANGE OF ORDER OF INTEGRATION(3)

METHOD – 13 : D.I. BY CHANGE OF VARIABLE IN CARTESIAN COORDINATES

METHOD – 14 : D.I. BY CHANGE OF VARIABLE IN POLAR COORDINATES(2)

METHOD – 7 : AREA BY DOUBLE INTEGRATION IN CARTESIAN COORDINATES(1)

METHOD – 8 : AREA BY DOUBLE INTEGRATION IN POLAR COORDINATES

METHOD – 4 : DOUBLE INTEGRALS AS VOLUMES(1)

METHOD – 9 : T.I. OVER GENERAL REGION IN CARTESIAN COORDINATES

METHOD – 10 : T.I. OVER GENERAL REGION IN CYLINDRICAL COORDINATES

METHOD – 11 : T.I. OVER GENERAL REGION IN SPHERICAL COORDINATES(1)

D.I. by change of Variables of integration in CC(Method-13)

Procedure to find double integral by change of Variable - CC

- If we want to find the value of given double integral by changing the variable of Integration, we will follow following steps:

1) Find transformation from given instruction if it is not given directly. i.e. relation between x, y(given variables) and u, v(new variables).
2) Express x & y in terms of u and v (if required).
3) Find Jacobian in **xy**-plane.
4) Convert given integral & Region in uv-plane using transformation.
5) Evaluate

$$\iint_R f(x,y)\,dx\,dy = \iint_G h(u,v)\cdot|J|\cdot du\,dv$$

$$J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$$

Procedure to find double integral by change of Variable - CC

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2) Express u & v in terms of x and y (if required).
3) Find Jacobian in **uv**-plane.
4) Convert given integral & Region in uv-plane using transformation.
5) Evaluate

$$J = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

$$\iint_R f(x, y)\, dx\, dy = \iint_G h(u, v) \frac{du\, dv}{|J|}$$

Method 13 ⇒ Example – 2

Evaluate the integral

$$\int_0^1 \int_0^{1-x} e^{\frac{y}{x+y}}\, dy\, dx \quad \text{by}$$

Changing the variables

$$x + y = u \ \ \& \ y = uv.$$

Solution:

We have $u = x + y \ \ \& \ \ y = uv$

$\therefore x = u - uv \ \ \& \ \ y = uv$

Now $J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$

$= \begin{vmatrix} 1-v & -u \\ v & u \end{vmatrix}$

$= u - uv + uv$

$\therefore J = u$

- Find Transformation
- Find Jacobian
- Find x & y, in terms of u & v.
- Find region & integral in uv-plane.
- Solve Double Integral.

Now $f(x,y) = e^{\left(\frac{y}{x+y}\right)}$

$\therefore h(u,v) = e^{\left(\frac{uv}{u}\right)} = e^{v}$

From the limits of integration,

$y = 0 \quad \Rightarrow uv = 0$

$\Rightarrow u = 0 \quad \text{or} \quad v = 0$

$y = 1 - x \quad \Rightarrow x + y = 1$

$\Rightarrow u = 1$

$x = 0 \quad \Rightarrow u - uv = 0$

$\Rightarrow u = 0 \quad \text{or} \quad 1 - v = 0$

$\Rightarrow u = 0 \quad \text{or} \quad v = 1$

$x = 1 \quad \Rightarrow u - uv = 1$

$\Rightarrow$ No conclusion

$$\therefore \int_0^1 \int_0^{1-x} e^{\frac{y}{x+y}}\, dy\, dx$$

$$= \iint_G h(u,v) \cdot |J| \cdot du\, dv$$

$$= \int_{v=0}^{1} \int_{u=0}^{1} e^{v} \cdot u \cdot du\, dv$$

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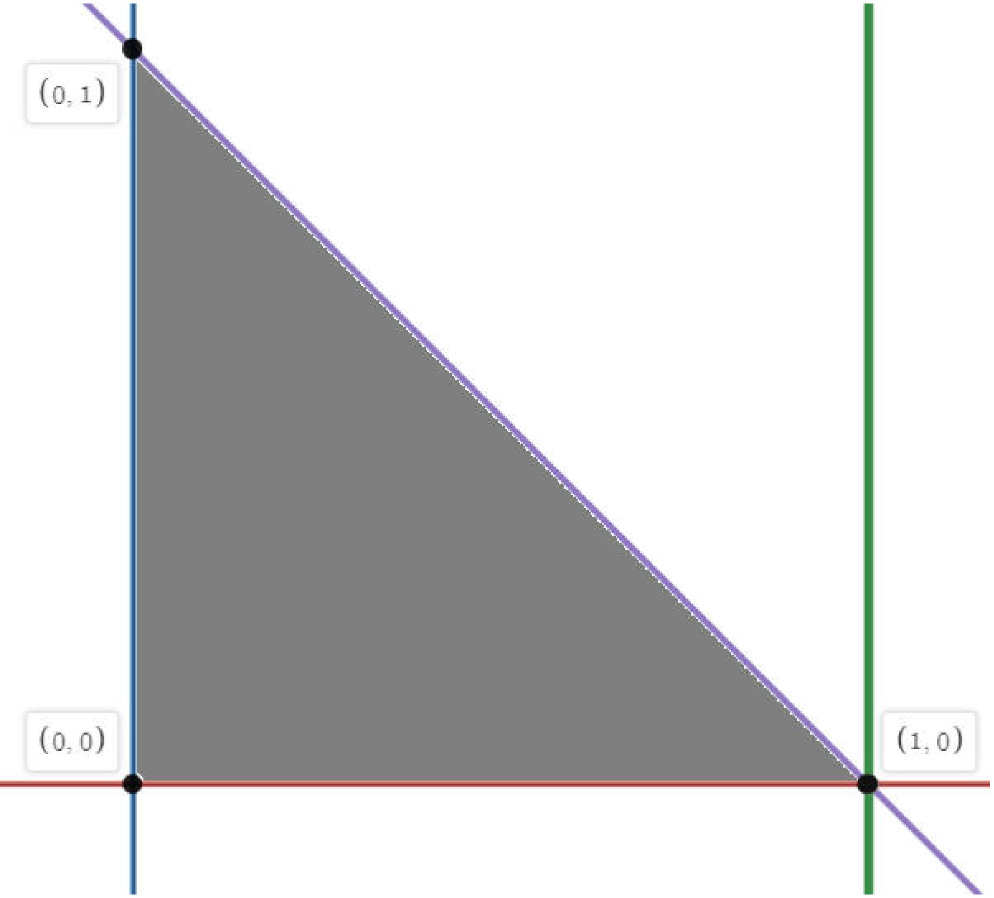

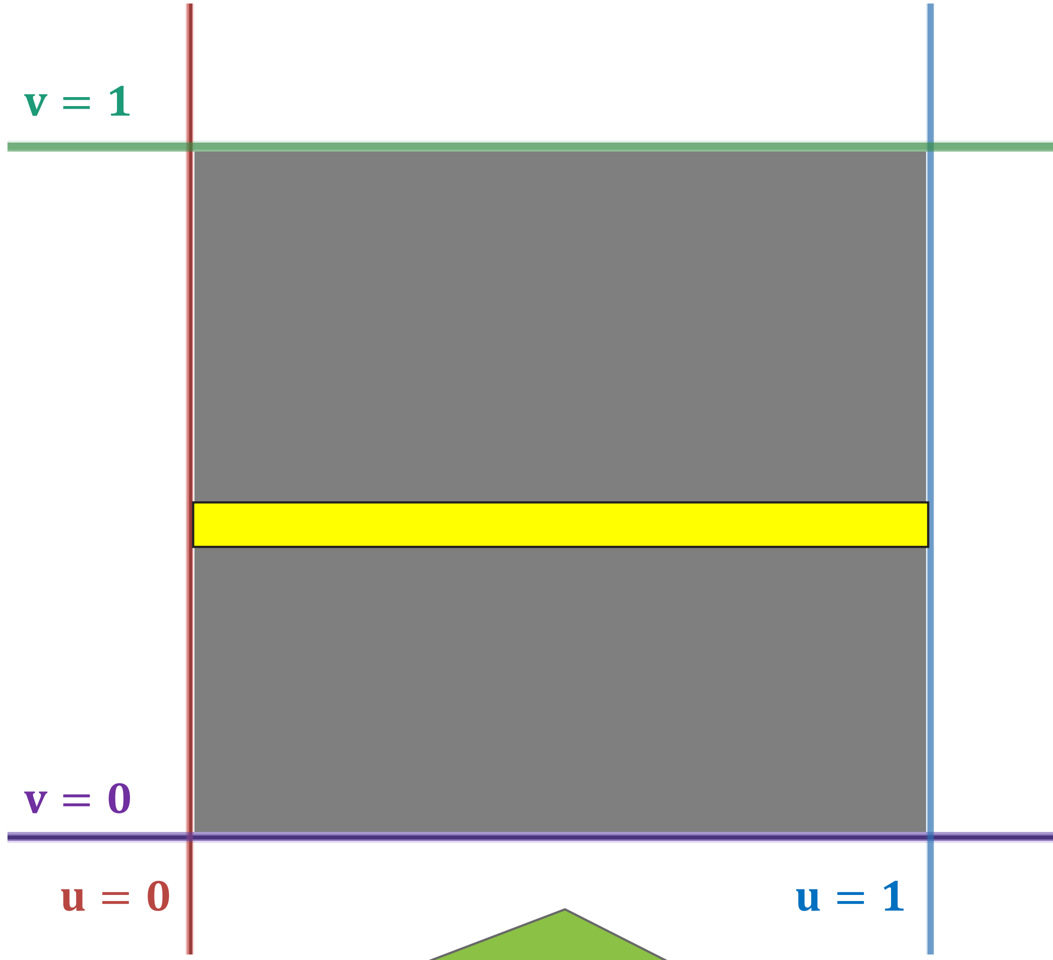

From XY plane to UV plane

Method 13 ⇒ Example – 5

Evaluate $\iint\limits_R (x+y)\,dA$, where R is the trapezoidal region with vertices $(0,0),(5,0),\left(\frac{5}{2},\frac{5}{2}\right)$ & $\left(\frac{5}{2},-\frac{5}{2}\right)$ using the transformations $x = 2u + 3v$ and $y = 2u - 3v$.

Solution:

We have $x = 2u + 3v$ & $y = 2u - 3v$

Now $J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$

$= \begin{vmatrix} 2 & 3 \\ 2 & -3 \end{vmatrix}$

$= -6 - 6$

$\therefore J = -12$

Now $f(x,y) = x + y$

$\therefore h(u,v) = 2u + 3v + 2u - 3v$

$= 4u$

- Find Transformation
- Find Jacobian
- Find x & y, in terms of u & v.
- Find region & integral in uv-plane.
- Solve Double Integral.

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From the Region **R**,

$$(x, y) = (0, 0)$$

$$\Rightarrow x = 0, y = 0$$

$$\Rightarrow 2u + 3v = 0,$$
$$2u - 3v = 0$$

$$\Rightarrow \boxed{u = 0}, \quad \boxed{v = 0}$$

$$(x, y) = (5, 0)$$

$$\Rightarrow x = 5, y = 0$$

$$\Rightarrow 2u + 3v = 5,$$
$$2u - 3v = 0$$

$$\Rightarrow \boxed{u = \frac{5}{4}}, \quad \boxed{v = \frac{5}{6}}$$

$$(x, y) = \left(\frac{5}{2}, \frac{-5}{2}\right)$$

$$\Rightarrow 2u + 3v = 5/2,$$
$$2u - 3v = -5/2$$

$$\Rightarrow u = 0, \quad v = \frac{5}{6}$$

Method 13 ⇒ Example – 5(continue)

$$\therefore \iint\limits_{R} (x+y)\, dA$$

$$= \iint\limits_{G} h(u,v) \cdot |\, J \,| \cdot du\, dv$$

$$= \int\limits_{v=0}^{5/6} \int\limits_{u=0}^{5/4} 4u \cdot 12 \cdot du\, dv$$

. . . . . .

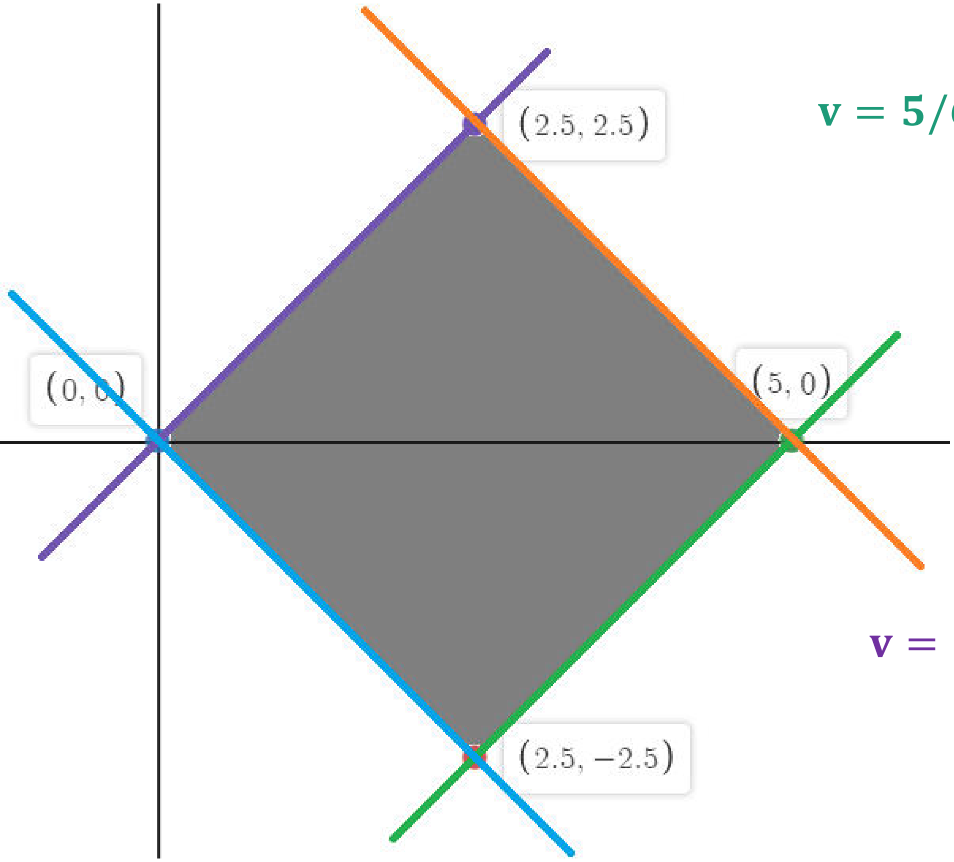
From the previous slide

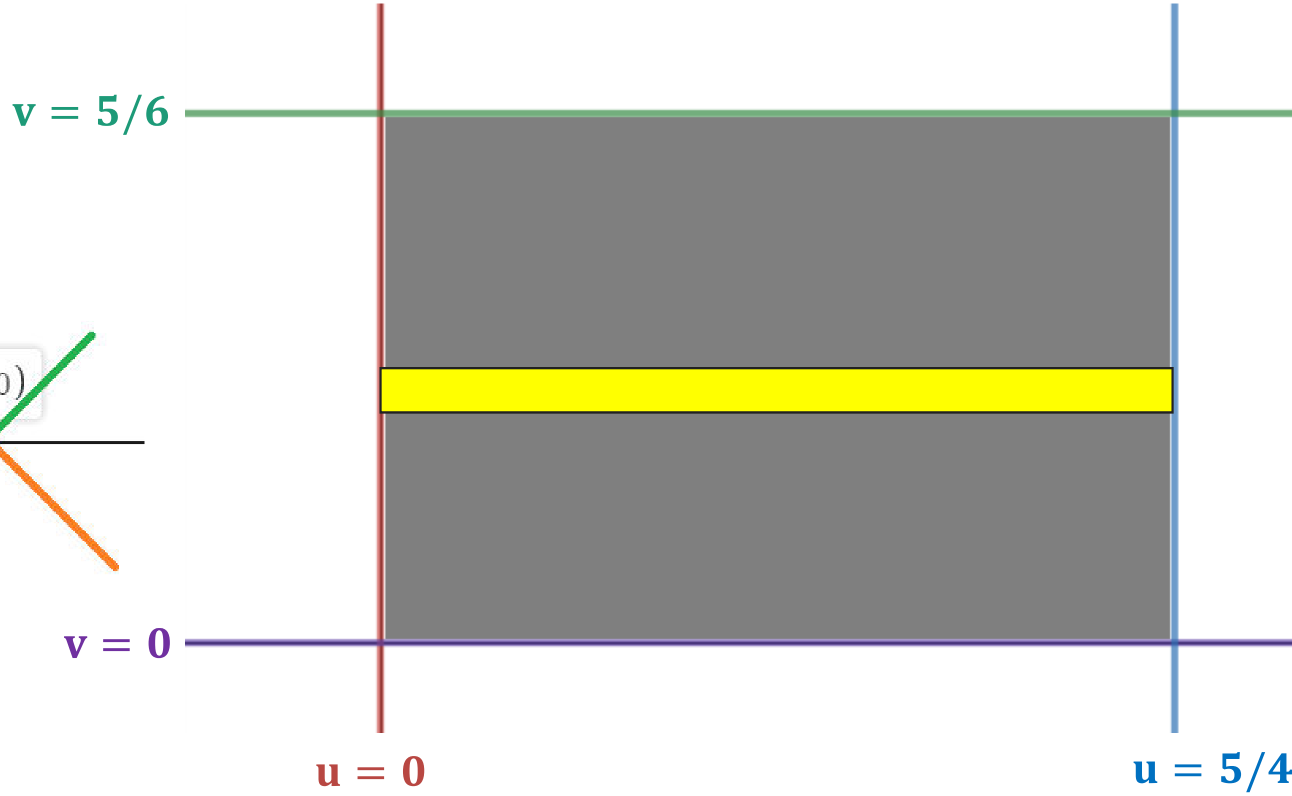
$h(u,v) = 4u$

$J = -12$

$u = 0, \qquad u = \dfrac{5}{4}$

$v = 0, \qquad v = \dfrac{5}{6}$





From XY plane to UV plane

Method 13 ⇒ Example – 9

Evaluate $\iint_R (x^2 - y^2)^2 \, dA$ over the area bounded by lines $|x| + |y| = 1$ using the transformations $x + y = u$ and $x - y = v$.

Solution:

We have $u = x + y \quad \& \quad v = x - y$

- Find **Transformation**
- Find **Jacobian**
- Find **x** & **y**, in terms of **u** & **v**.
- Find **region** & **integral** in uv-plane.
- **Solve** Double Integral.

Now $J = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$

$= \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$

$= -1 - 1$

$\therefore J = -2$

Now $f(x, y) = (x^2 - y^2)^2$

$\therefore f(x, y) = \{ (x + y)(x - y) \}^2$

$\therefore h(u, v) = u^2 v^2$

Method 13 ⇒ Example – 9(continue)

From the Region **R** bounded by lines $|x| + |y| = 1$,

$x + y = 1 \quad \Rightarrow u = 1$

$x - y = 1 \quad \Rightarrow v = 1$

$-x + y = 1 \quad \Rightarrow x - y = -1 \quad \Rightarrow v = -1$

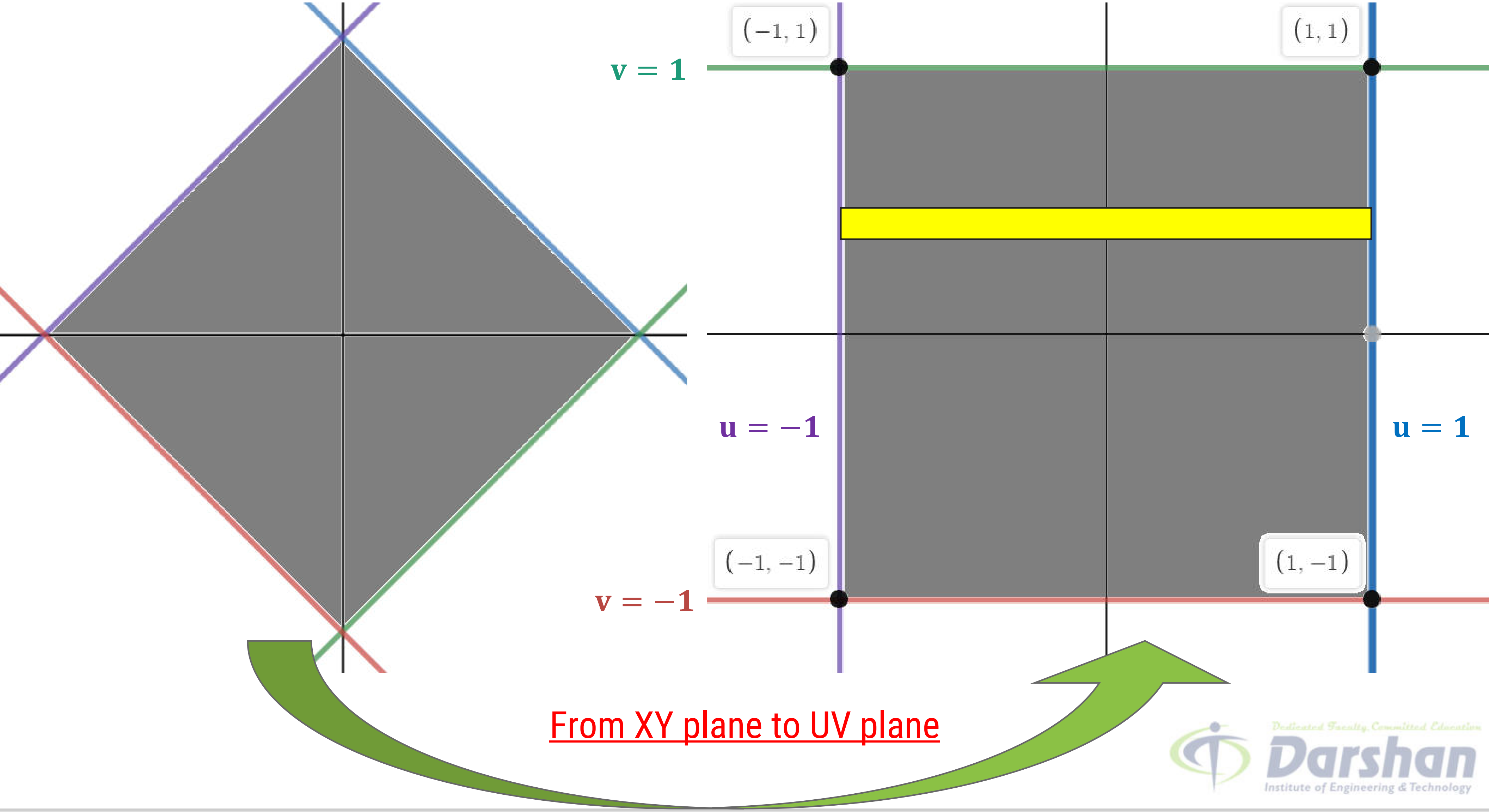
$-x - y = 1 \quad \Rightarrow x + y = -1 \quad \Rightarrow u = -1$

$$\therefore \iint_R (x^2 - y^2)^2 \, dA$$

$$= \iint_G h(u,v) \frac{du\,dv}{|J|}$$

$$= \int_{v=-1}^{1} \int_{u=-1}^{1} u^2 v^2 \frac{du\,dv}{2}$$

……



From XY plane to UV plane

Method 13 $\Rightarrow$ Example – 3

Evaluate $\iint\limits_R (x^2 + y^2)dA$, by changing the variables, where R is the region lying in the first quadrant and bounded by the hyperbolas

$x^2 - y^2 = 9,\quad x^2 - y^2 = 1,\quad xy = 2$ and $xy = 4.$

Solution:

Here transformation is not given directly.

So from the instruction we will take $u = x^2 - y^2$ and $v = xy.$

 $u = x^2 - y^2$ and $v = xy.$

$$\text{Now } J = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

$$= \begin{vmatrix} 2x & -2y \\ y & x \end{vmatrix}$$

$$= 2x^2 + 2y^2$$

$$\therefore J = 2(x^2 + y^2)$$

$$\text{Now } f(x,y) = x^2 + y^2$$

From the Region **R**,

$$x^2 - y^2 = 9 \Rightarrow u = 9$$

$$x^2 - y^2 = 1 \Rightarrow u = 1$$

$$xy = 2 \Rightarrow v = 2$$

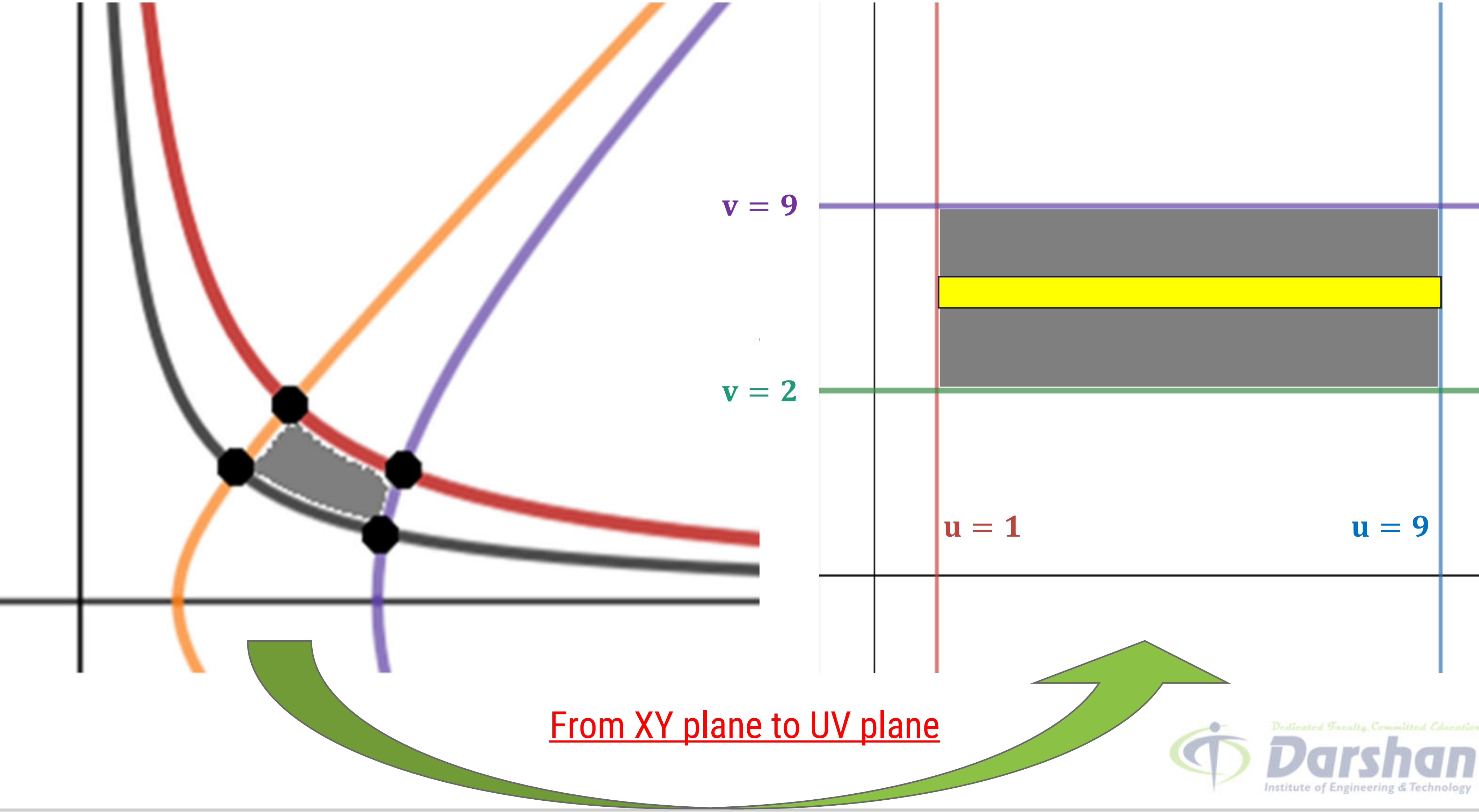
$$xy = 4 \Rightarrow v = 4$$

$$\therefore \iint_R (x^2 + y^2)\, dA$$

$$= \iint_G h(u,v) \frac{du\, dv}{|J|}$$

$$= \int_{v=2}^{4} \int_{u=1}^{9} \frac{1}{2}\, du\, dv$$

……



From XY plane to UV plane

D.I. by change of Variables of integration in PC(Method-14)

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Procedure to find double integral by change of Variable - PC

- If we want to find the value of given double integral by changing the variable of Integration, we will follow following steps:

1) Use transformation $x = r\cos\theta$ and $y = r\sin\theta$.
   i.e. relation between x, y(given variables) and r, θ(new variables).
2) Draw the region in xy-plane.
3) Convert given integral & Region in $r\theta$-plane using transformation.
4) Evaluate

$$\iint_R f(x,y)\,dx\,dy = \iint_G h(r,\theta)\cdot \mathbf{r}\cdot dr\,d\theta$$

Method 14 ⇒ Example – 1

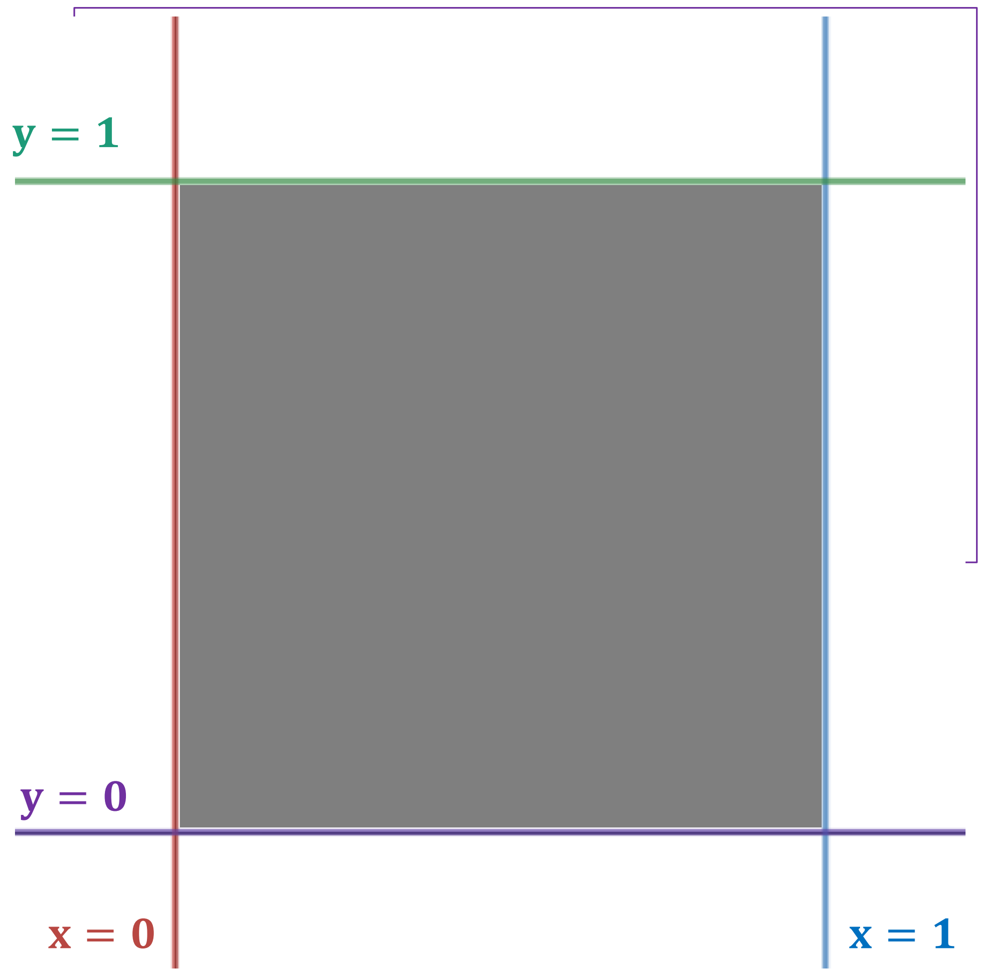
By changing into polar coordinates, evaluate $\int_0^1 \int_0^1 dx\, dy$.

Solution:

Let us take $x = r\cos\theta$ and $y = r\sin\theta$.

Form the limits of integral we have,

$x = 0, \quad x = 1 \quad \&$
$y = 0, \quad y = 1.$



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Method 14 $\Rightarrow$ Example – 1(continue)

From the Region **R**,

$x = 0 \quad \Rightarrow r\cos\theta = 0 \quad \Rightarrow r = 0 \text{ or } \cos\theta = 0 \quad \Rightarrow r = 0 \text{ or } \theta = \pi/2$

$x = 1 \quad \Rightarrow r\cos\theta = 1 \quad \Rightarrow r = 1/\cos\theta \quad \Rightarrow r = \sec\theta$

$y = 0 \quad \Rightarrow r\sin\theta = 0 \quad \Rightarrow r = 0 \text{ or } \sin\theta = 0 \quad \Rightarrow r = 0 \text{ or } \theta = 0$

$y = 1 \quad \Rightarrow r\sin\theta = 1 \quad \Rightarrow r = 1/\sin\theta \quad \Rightarrow r = \operatorname{cosec}\theta$

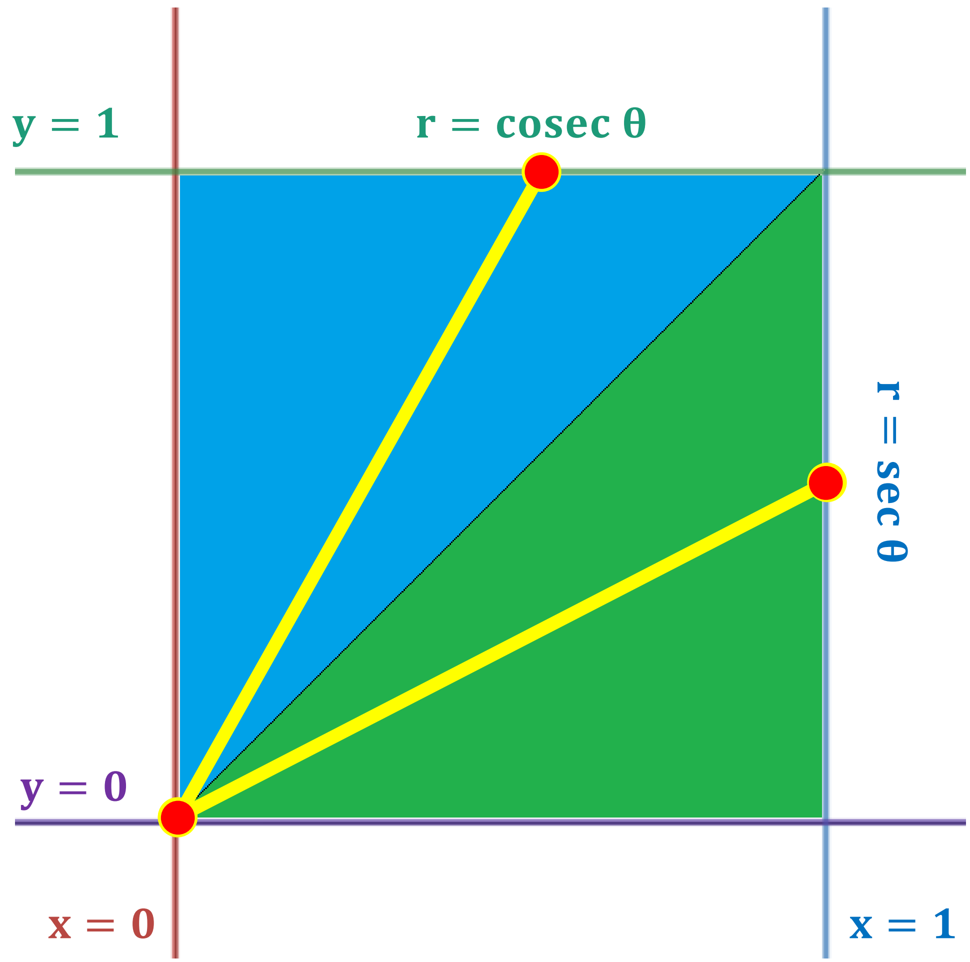
And $f(x, y) = 1 \ \Rightarrow f(r, \theta) = 1$

Method 14 ⇒ Example – 1(continue)

$$\int_0^1 \int_0^1 dx\,dy = \iint_G h(r,\theta)\cdot r \cdot dr\,d\theta$$

$$= \iint_{G_1} r\ dr\,d\theta + \iint_{G_2} r\ dr\,d\theta$$

$$= \int_0^{\pi/4} \int_0^{\sec\theta} r\ dr\,d\theta + \int_{\pi/4}^{\pi/2} \int_0^{\operatorname{cosec}\theta} r\ dr\,d\theta$$



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Evaluate the integral

$$\int_0^\infty \int_0^\infty \frac{1}{(1+x^2+y^2)^2}\, dx\, dy.$$

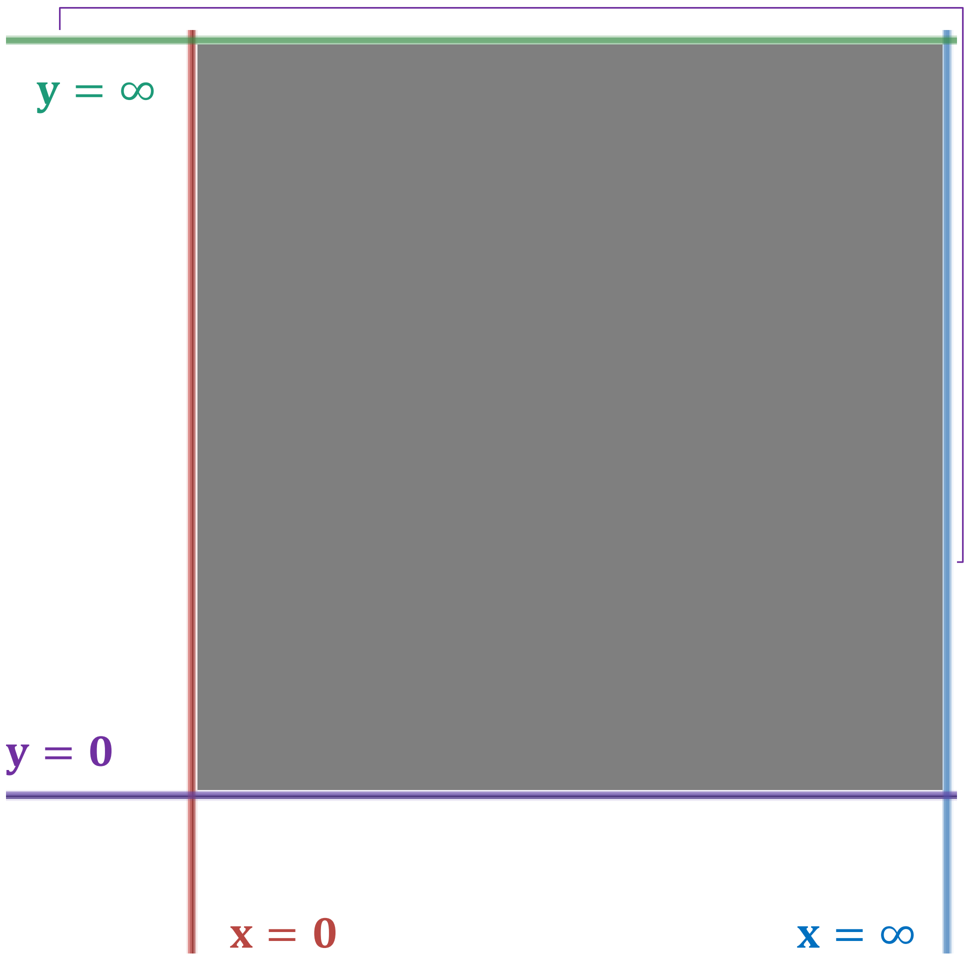
Solution: S-19(3marks)

Let us take $x = r\cos\theta$ and $y = r\sin\theta$.

Form the limits of integral we have,

$x = 0, \quad x = \infty \quad \&$

$y = 0, \quad y = \infty.$



Method 14 ⇒ Example – 3(continue)

From the Region **R**,

$x = 0 \quad \Rightarrow r\cos\theta = 0 \quad \Rightarrow r = 0$ or $\cos\theta = 0 \quad \Rightarrow r = 0$ or $\theta = \pi/2$

$x = \infty \quad \Rightarrow r\cos\theta = \infty \quad \Rightarrow r = \infty$

$y = 0 \quad \Rightarrow r\sin\theta = 0 \quad \Rightarrow r = 0$ or $\sin\theta = 0 \quad \Rightarrow r = 0$ or $\theta = 0$

$y = \infty \quad \Rightarrow r\sin\theta = \infty \quad \Rightarrow r = \infty$

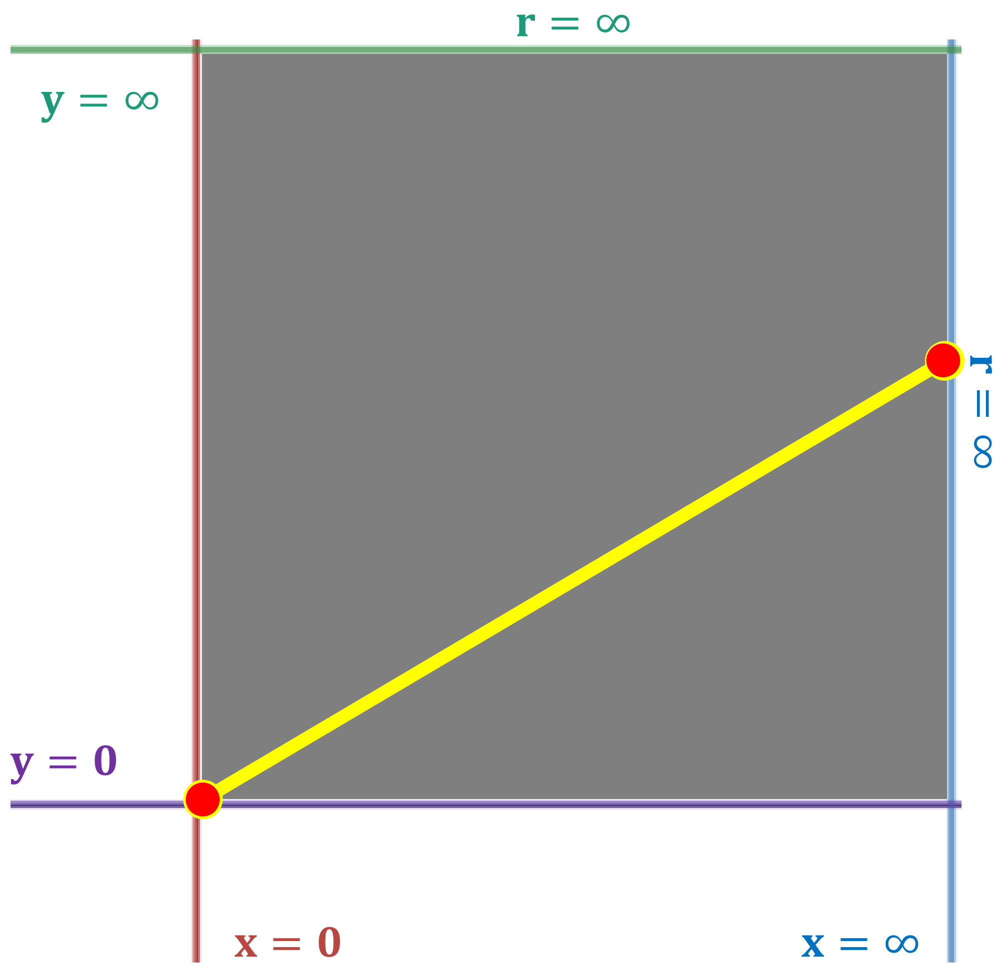
And $f(x,y) = \dfrac{1}{(1+x^2+y^2)^2} \quad \Rightarrow f(r,\theta) = \dfrac{1}{(1+r^2)^2}$

Method 14 ⇒ Example – 3(continue)

$$\int_0^1 \int_0^1 dx\,dy = \iint_G h(r,\theta)\cdot \mathbf{r} \cdot dr\,d\theta$$

$$= \int_0^{\pi/2} \int_0^{\infty} \frac{1}{(1+r^2)^2}\, r\,dr\,d\theta$$

……



By changing into polar co—ordinates, evaluate the integral

$$\int_0^{2a}\int_0^{\sqrt{2ax-x^2}} (x^2+y^2)\, dy\, dx.$$

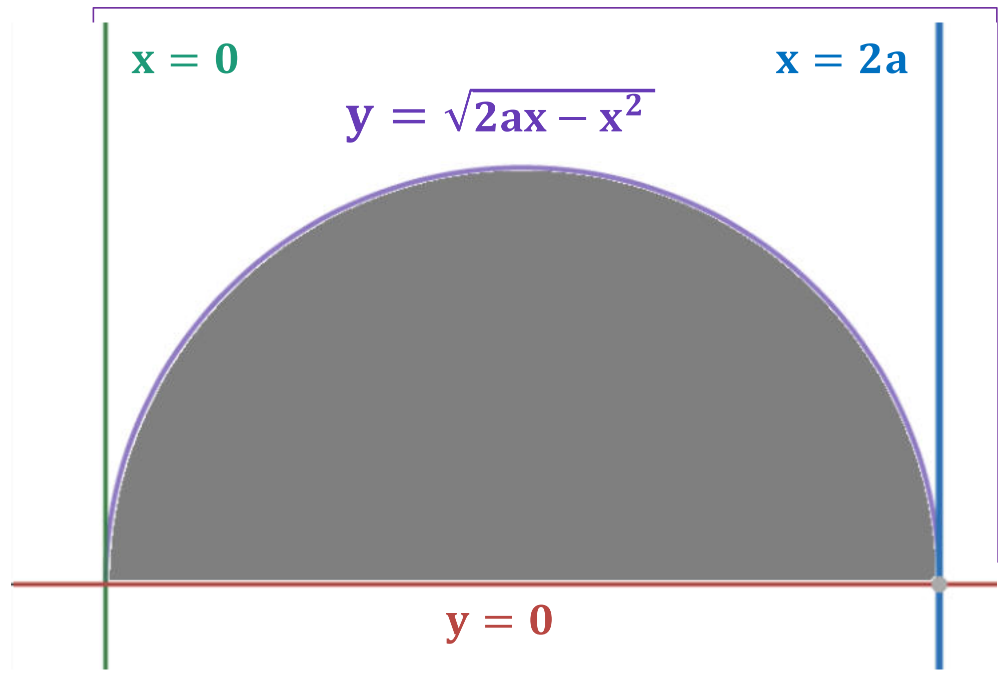
Solution:

Let us take x = r cos θ and y = r sin θ.

Form the limits of integral we have,

$y = 0,\quad y = \sqrt{2ax - x^2}\quad \&$

$x = 0,\quad x = 2a.$



Method 14 $\Rightarrow$ Example – 7(continue)

From the Region **R**,

$y = 0 \quad \Rightarrow r\sin\theta = 0 \quad \Rightarrow r = 0$ or $\sin\theta = 0 \quad \Rightarrow r = 0$ or $\theta = 0$

$x = 0 \quad \Rightarrow r\cos\theta = 0 \quad \Rightarrow r = 0$ or $\cos\theta = 0 \quad \Rightarrow r = 0$ or $\theta = \pi/2$

$x = 2a \quad \Rightarrow r\cos\theta = 2a \quad \Rightarrow r = 2a\sec\theta$

$y = \sqrt{2ax - x^2} \quad \Rightarrow r = 2a\cos\theta$

$$
\begin{aligned}
y &= \sqrt{2ax - x^2} \\
y^2 &= 2ax - x^2 \\
r^2\sin^2\theta &= 2ar\cos\theta - r^2\cos^2\theta \\
r^2 &= 2ar\cos\theta
\end{aligned}
$$

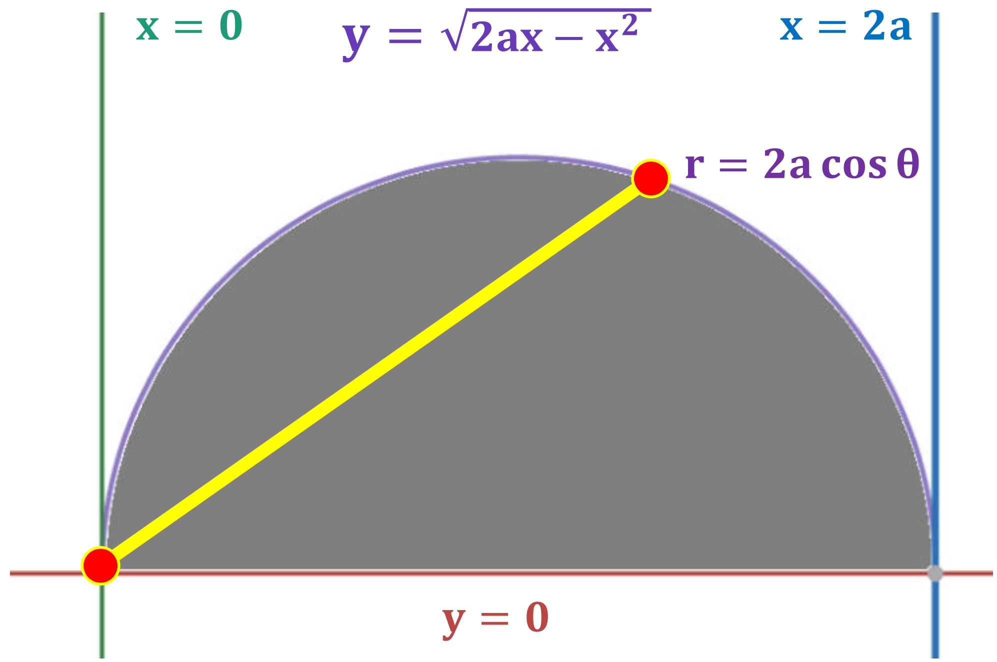
And $f(x, y) = x^2 + y^2 \quad \Rightarrow f(r, \theta) = r^2$

Method 14 $\Rightarrow$ Example – 7(continue)

$$\int_0^{2a}\int_0^{\sqrt{2ax-x^2}} (x^2+y^2)\,dy\,dx$$

$$= \iint_G h(r,\theta)\cdot \mathbf{r}\cdot dr\,d\theta$$

$$= \int_0^{\pi/2}\int_0^{2a\cos\theta} r^2\cdot r\cdot dr\,d\theta$$



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For Reference:

$$\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \int_0^{\frac{\pi}{2}} \cos^n x \, dx = I_n \quad \Rightarrow I_n = \frac{n-1}{n} I_{n-2} \quad ; \text{where} \quad I_0 = \frac{\pi}{2}, \quad I_1 = 1$$

$$\int_0^{\frac{\pi}{2}} \sin^6 x \, dx = I_6 \quad \Rightarrow I_6 = \frac{5}{6} I_4 = \frac{5}{6}\,\frac{3}{4} I_2 = \frac{15}{24}\,\frac{1}{2} I_0 = \frac{15}{48}\,\frac{\pi}{2}$$

$$\Rightarrow I_6 = \frac{5}{6}\,\frac{3}{4}\,\frac{1}{2}\,\frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \cos^7 x \, dx = I_7 \quad \Rightarrow I_7 = \frac{6}{7} I_5 = \frac{6}{7}\,\frac{4}{5} I_3 = \frac{24}{35}\,\frac{2}{3} I_1 = \frac{48}{105}\,1$$

$$\Rightarrow I_7 = \frac{6}{7}\,\frac{4}{5}\,\frac{2}{3}\,1$$

Method 14 ⇒ Example – 9

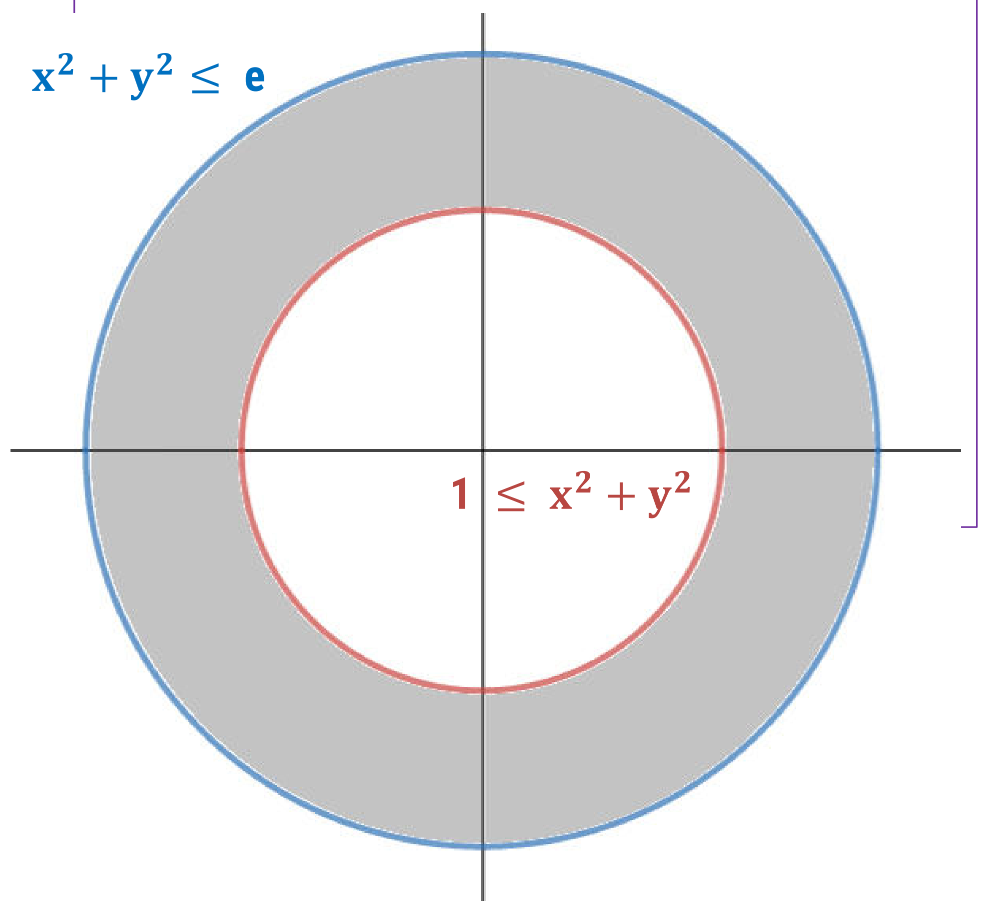
Integrate $f(x,y) = \dfrac{\ln\,(x^2+y^2)}{\sqrt{x^2+y^2}}$ over the region $1 \le x^2+y^2 \le e$ by changing into polar coordinates.

Solution:

Let us take $x = r\cos\theta$ and $y = r\sin\theta$.

Form the region we have,

$1 \le x^2+y^2$ & $x^2+y^2 \le e$



Method 14 ⇒ Example – 9(continue)

From the Region **R**,

$$x^2 + y^2 = 1 \qquad \Rightarrow \boxed{r = 1}$$

$$x^2 + y^2 = e \qquad \Rightarrow \boxed{r = \sqrt{e}}$$

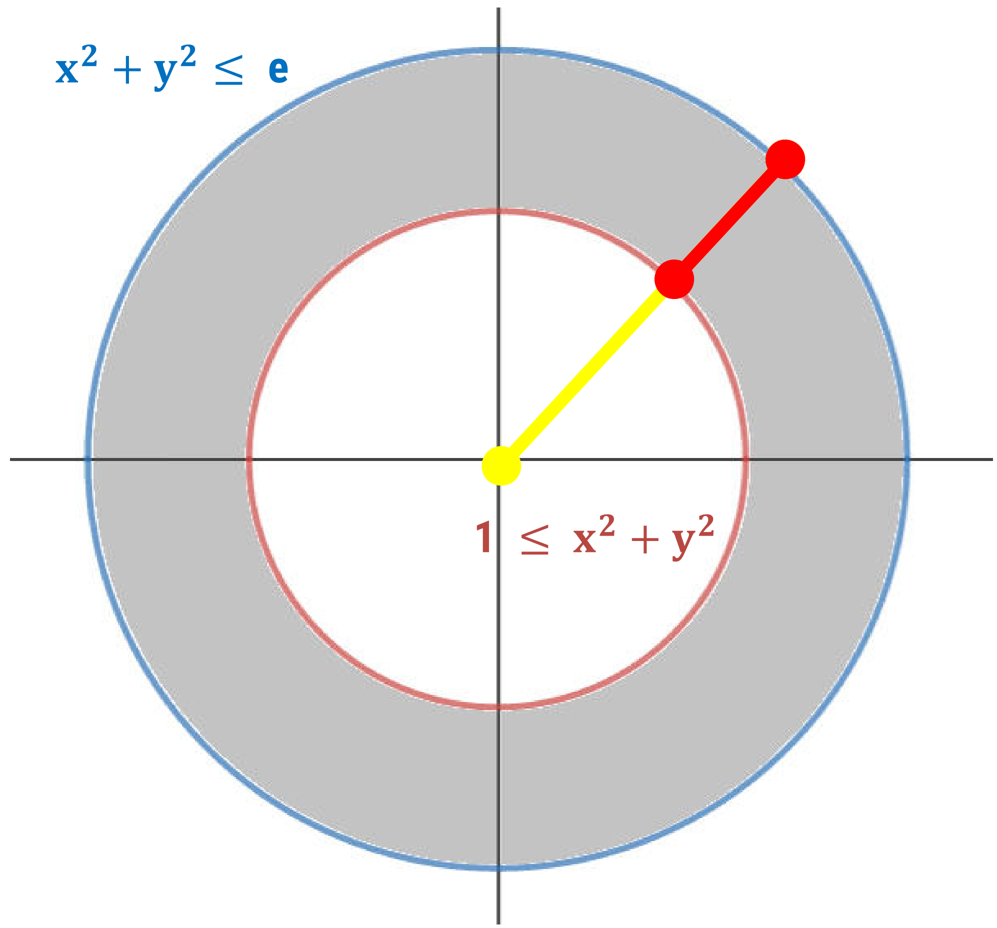
$$\text{And } f(x, y) = \frac{\ln\ (x^2+y^2)}{\sqrt{x^2 + y^2}} \quad \Rightarrow \boxed{f(r, \theta) = \frac{\ln(r^2)}{r}}$$

Method 14 ⇒ Example – 9(continue)

$$\iint_R \left(\frac{\ln\ (x^2+y^2)}{\sqrt{x^2+y^2}} \right) dA$$

$$= \iint_G h(r,\theta) \cdot \mathbf{r} \cdot dr\, d\theta$$

$$= \int_0^{2\pi} \int_1^{\sqrt{e}} \frac{\ln(r^2)}{r} \cdot r \cdot dr\, d\theta$$



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Area by Double integration in CC(Method-7)

Area by Double integration in CC(Method-7)

- The **area** of a closed, bounded plane region *R* is

$$\text{Area} = \iint\limits_{R} dA$$

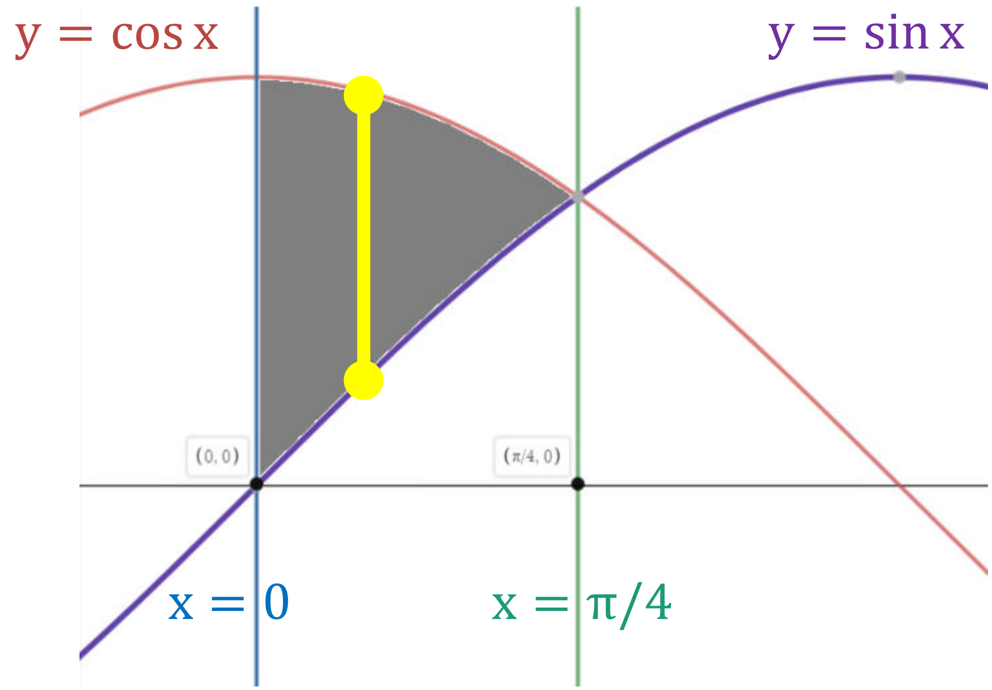
$$\text{Area} = \iint\limits_{R} dx\, dy$$

$$\text{Area} = \iint\limits_{R} dy\, dx$$

Method 7 ⇒ Example – 2

Find the area of the region bounded by the curves $y = \sin x$, $y = \cos x$ and the lines $x = 0$ and $x = \pi/4$.

Solution: S-19 (4 marks)



We have

$y = \sin x, \quad y = \cos x$ &

$x = 0$ and $x = \dfrac{\pi}{4}$

$$\therefore \text{Area} = \iint_R dA = \int_0^{\frac{\pi}{4}} \int_{\sin x}^{\cos x} \mathbf{dy}\, dx$$

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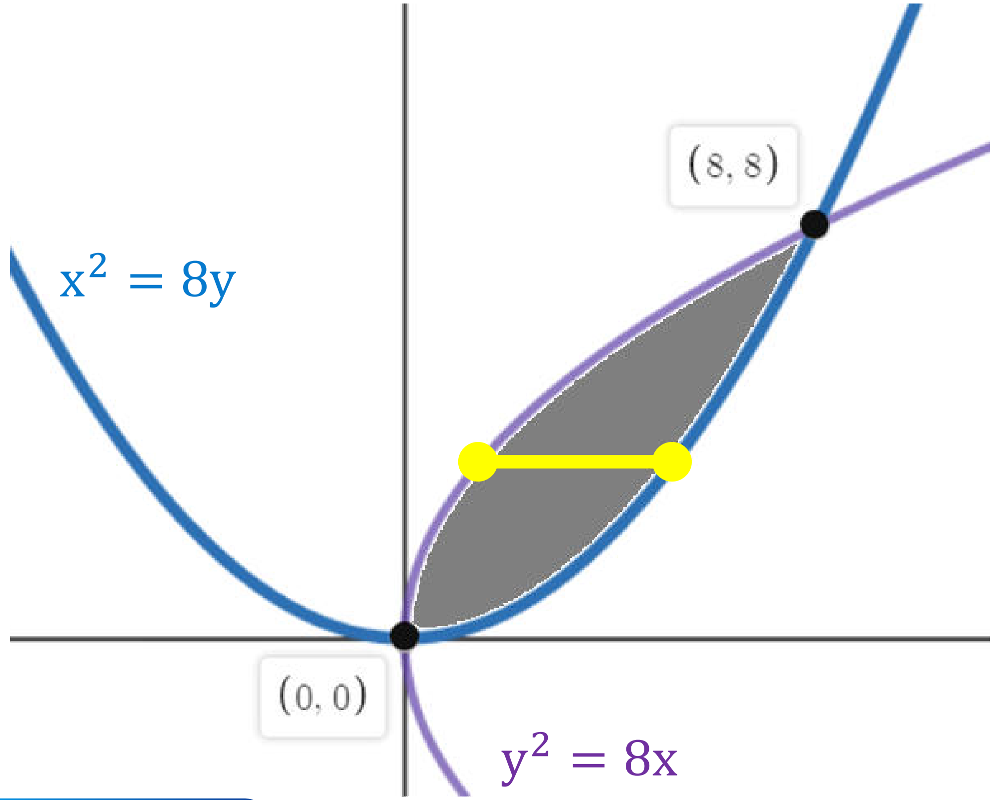
Method 7 ⇒ Example – 3

Using the double integration, find the area of the region common to the parabolas $y^2 = 8x$ and $x^2 = 8y$.



Solution:

We have

$y^2 = 8x$ and $x^2 = 8y$

$$\therefore \text{Area} = \iint_R dA = \int_0^8 \int_{y^2/8}^{\sqrt{8y}} \mathbf{dx}\, dy$$

. . . . . .

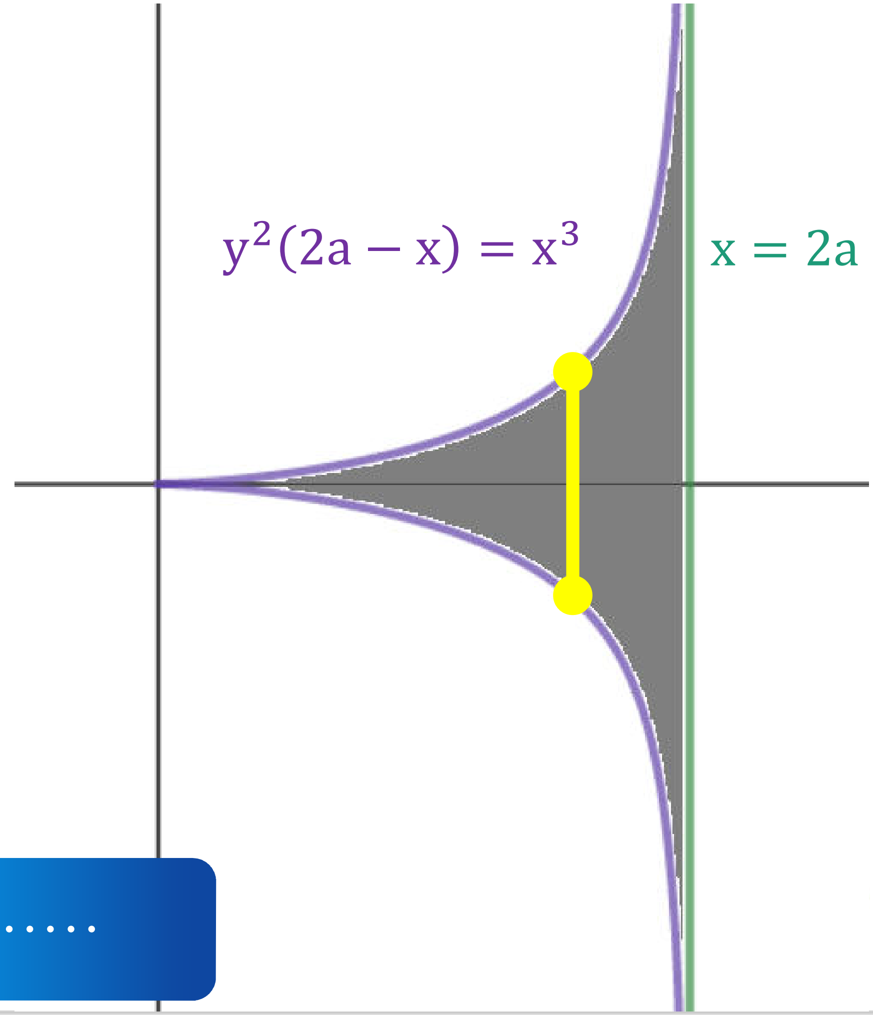
Method 7 ⇒ Example – 5

Find the area of region included between the curve $y^2(2a-x)=x^3$ and its asymptote.

Solution:

We have $y^2(2a-x)=x^3$ & $x=2a$

$$\therefore \text{Area} = \iint_R dA = \int_0^{2a} \int_{-\sqrt{\frac{x^3}{2a-x}}}^{+\sqrt{\frac{x^3}{2a-x}}} \mathbf{dy}\, dx$$



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Method 7 ⇒ Example – 5(continue)

Sol^n-5
(M-7)

$$\text{Area} = \int_0^{2a} \int_{-\sqrt{\frac{x^3}{2a-x}}}^{\sqrt{\frac{x^3}{2a-x}}} 1 \, dy \, dx$$

$$= \int_0^{2a} 2\sqrt{\frac{x^3}{2a-x}} \, dx \quad \text{——— (1)}$$

Let $x = 2a \sin^2\theta \Rightarrow dx = 2a \, 2 \sin\theta \cos\theta d\theta$

& If $x = 0 \Rightarrow 2a \sin^2\theta = 0 \Rightarrow \theta = 0$

$x = 2a \Rightarrow 2a \sin^2\theta = 2a \Rightarrow \theta = \pi/2$

Method 7 ⇒ Example – 5(continue)

∴ From eq^n (1),

$$\text{Area} = 2\int_0^{\pi/2}\left(\sqrt{\frac{8a^3\sin^6\theta}{2a-2a\sin^2\theta}}\;\frac{4a\sin\theta}{\cos\theta}\right)d\theta$$

$$= 8a\int_0^{\pi/2}\left(\sqrt{\frac{8a^3\sin^6\theta}{2a\cos^2\theta}}\;\frac{\sin\theta}{\cos\theta}\right)d\theta$$

$$= 8a\int_0^{\pi/2}\frac{\sqrt{4a^2}\sqrt{\sin^6\theta}\;\sin\theta\cos\theta\, d}{\sqrt{\cos^2\theta}}$$

$$= 16a^2\int_0^{\pi/2}\sin^4\theta\; d\theta$$

$$= 16a^2 I_4$$

$$= 16a^2\;\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

$$\text{Area} = \boxed{3\pi a^2}$$

Answer

Area by Double integration in PC(Method-8)

Area by Double integration in PC(Method-8)

- The **area** of a closed, bounded plane region *R* is

$$\text{Area} = \iint\limits_R dA$$

$$\text{Area} = \iint\limits_R r\, dr\, d\theta$$

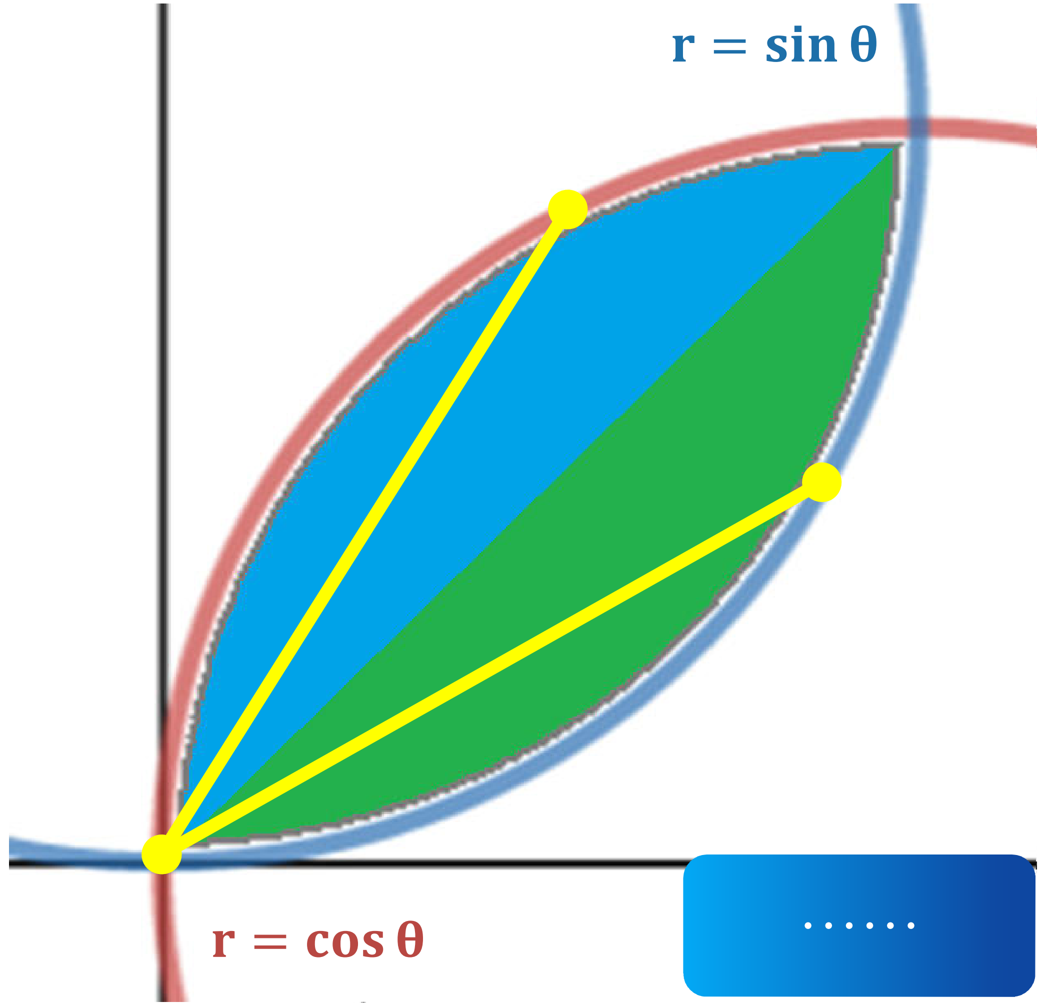
Method 8 ⇒ Example – 2

Find the area common to both of the circles $r = \cos\theta$ and $r = \sin\theta$.

Solution:

We have $r = \cos\theta$ and $r = \sin\theta$

$$\therefore \text{Area} = \iint_R dA = \int_0^{\pi/4}\int_0^{\sin\theta} r\,dr\,d\theta + \int_{\pi/4}^{\pi/2}\int_0^{\cos\theta} r\,dr\,d\theta$$



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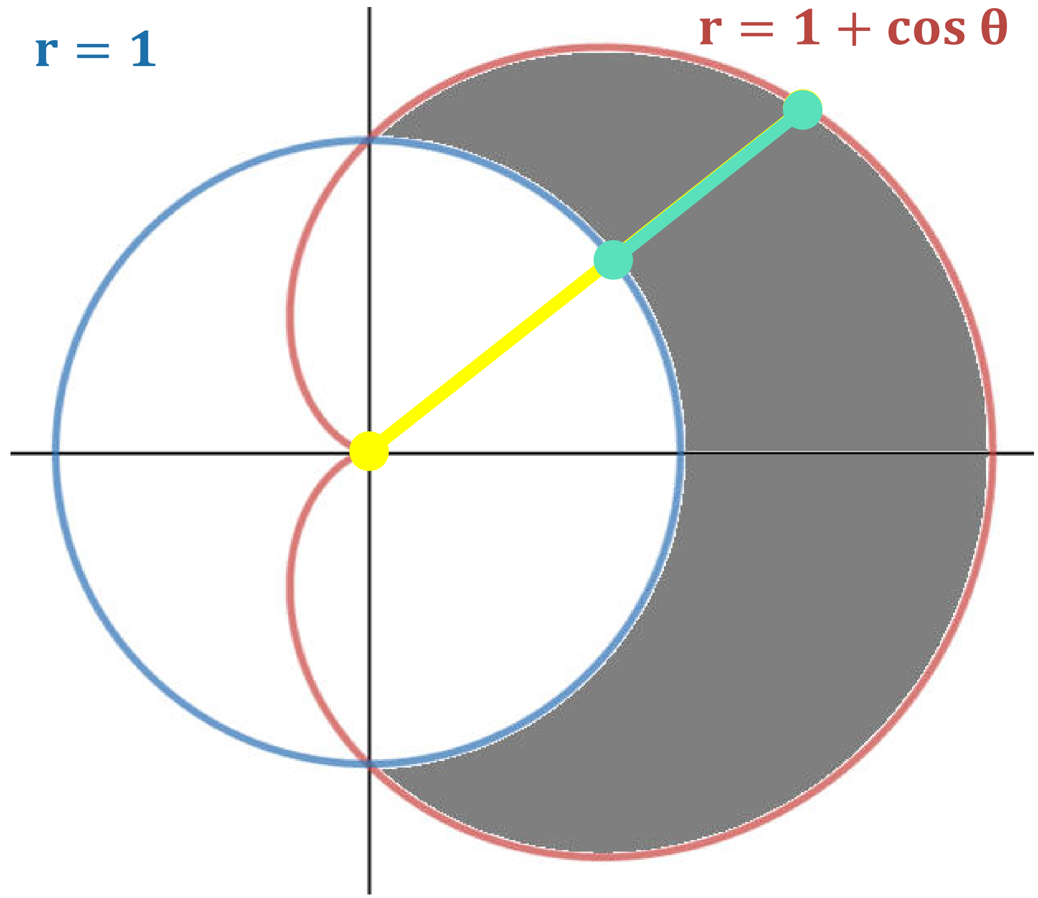
Find the area of the region that lies inside the cardioid $r = 1 + \cos\theta$ and outside the circle $r = 1$ by double integration.

Solution:

We have $r = 1 + \cos\theta$ and $r = 1$

$$\therefore \text{Area} = \iint_R dA = \int_{-\pi/2}^{\pi/2} \int_{1}^{1+\cos\theta} r\, \mathbf{dr}\, d\theta$$



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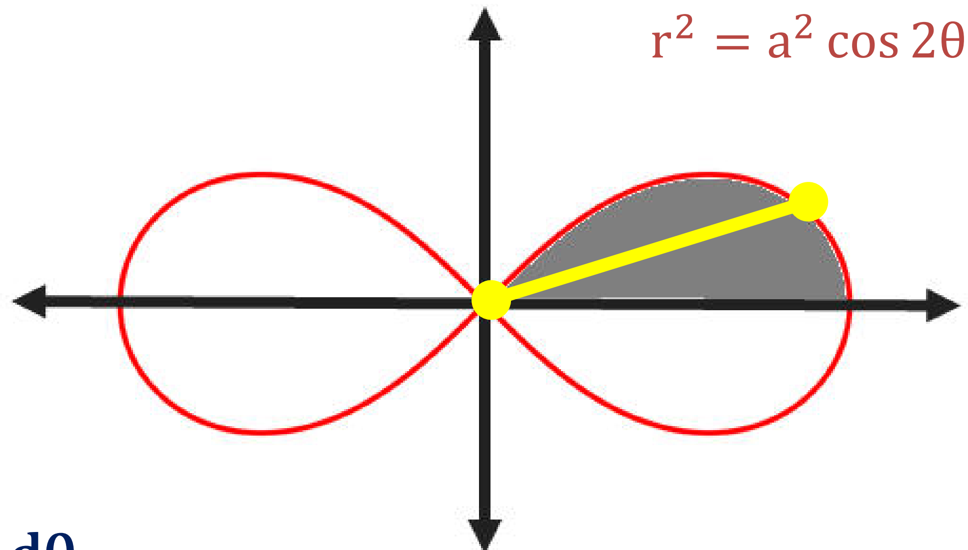
Find the total area enclosed by lemniscate $r^2 = a^2\cos 2\theta$.

Solution:

We have $r^2 = a^2\cos 2\theta$



$$\therefore \text{Area} = \iint_R dA = \mathbf{24}\int_{-\pi/4}^{\pi/4}\int_{0}^{\sqrt{a^2\cos 2\theta}} \mathbf{r\,dr\,d\theta}$$

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